

Li-Fi Based Data Transmission

Wireless Communication Using Visible Light



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Course

FWPM - Internet of Things

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Introduction & Problem

Understanding RF limitations in wireless communication and exploring Li-Fi as an innovative alternative for secure, high-speed local data transmission using visible light technology.

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System Design & Architecture

Complete hardware setup featuring Arduino Uno boards, LED transmitters, and solar panel receivers, integrated into a robust system architecture for reliable light-based communication.

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Implementation, Testing and Business Model

Data encoding and transmission logic using ASCII to binary conversion, precise threshold calibration methodology for accurate signal detection and possible business model.

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Challenges & Future Scope

Comprehensive testing results, performance analysis, identified challenges with solutions, and a strategic roadmap for future improvements and system optimization.

01

Introduction & Problem

Understanding RF limitations and the Li-Fi alternative
for secure, high-speed local communication



RF Challenges



Li-Fi Solution



Applications

Problem Statement

Traditional wireless systems face critical limitations in modern environments



RF Spectrum Congestion

Too many wireless devices share the same radio frequency spectrum, causing severe interference, signal degradation, and network overload in dense environments.



Reduced Speed in Dense Areas

Available bandwidth is shared between numerous users and devices, resulting in dramatically reduced communication speeds and increased latency.



Security Risks

RF signals can penetrate through walls and physical barriers, making them accessible outside the intended room and creating significant security vulnerabilities.



RF-Sensitive Environments

Hospitals, aircraft cabins, and industrial systems often restrict or prohibit RF communication due to potential interference with sensitive electronic equipment.



Critical Need

An alternative communication channel that avoids RF spectrum entirely while providing secure, high-speed local data transmission

Proposed Solution: Li-Fi Communication

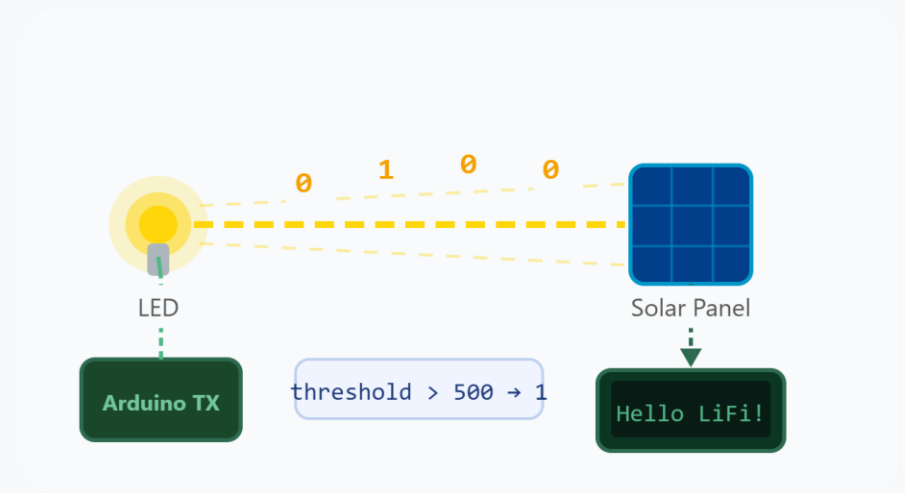
Using visible light as a secure, high-speed alternative to RF-based wireless systems

💡 How Li-Fi Works

- 1 LED ON represents binary 1
- 2 LED OFF represents binary 0
- 3 Characters converted to **8-bit ASCII binary** before transmission
- 4 Receiver uses **threshold-based detection** to decode light signals

✅ Key Advantages

- 🛡️ No RF interference
- 🔒 Physically contained
- ⚡ High-speed potential
- 👁️ Visible light spectrum



Binary Encoding

LED State

Binary Value

● LED ON

1

● LED OFF

0

Use Cases & Applications

Li-Fi technology enables secure communication across diverse environments



Hospitals & Medical Medical

Supports local data transmission without RF interference near sensitive medical equipment, ensuring patient safety and reliable communication.

✓ No RF interference with MRI, CT scans



Aircraft Connectivity Connectivity

Uses cabin LED lights for passenger communication while eliminating RF interference concerns with aircraft navigation systems.

✓ Safe during all flight phases



Industrial Automation

Reliable communication in high-electromagnetic-interference environments where RF signals may be unstable or unreliable.

✓ Immune to electromagnetic noise



Secure Rooms

Light-based signals stay contained within the room since visible light cannot pass through walls, providing natural physical security.

✓ Physical barrier = security



Educational IoT Prototype Prototype

Demonstrates core IoT concepts including sensors, embedded systems, connectivity, data encoding, and decoding in a simple end-to-end setup perfect for learning and experimentation.

✓ Hands-on learning

✓ Complete IoT pipeline

System Design & Architecture

Complete hardware setup, system flow,
and component integration for the Li-Fi prototype



Hardware



Architecture

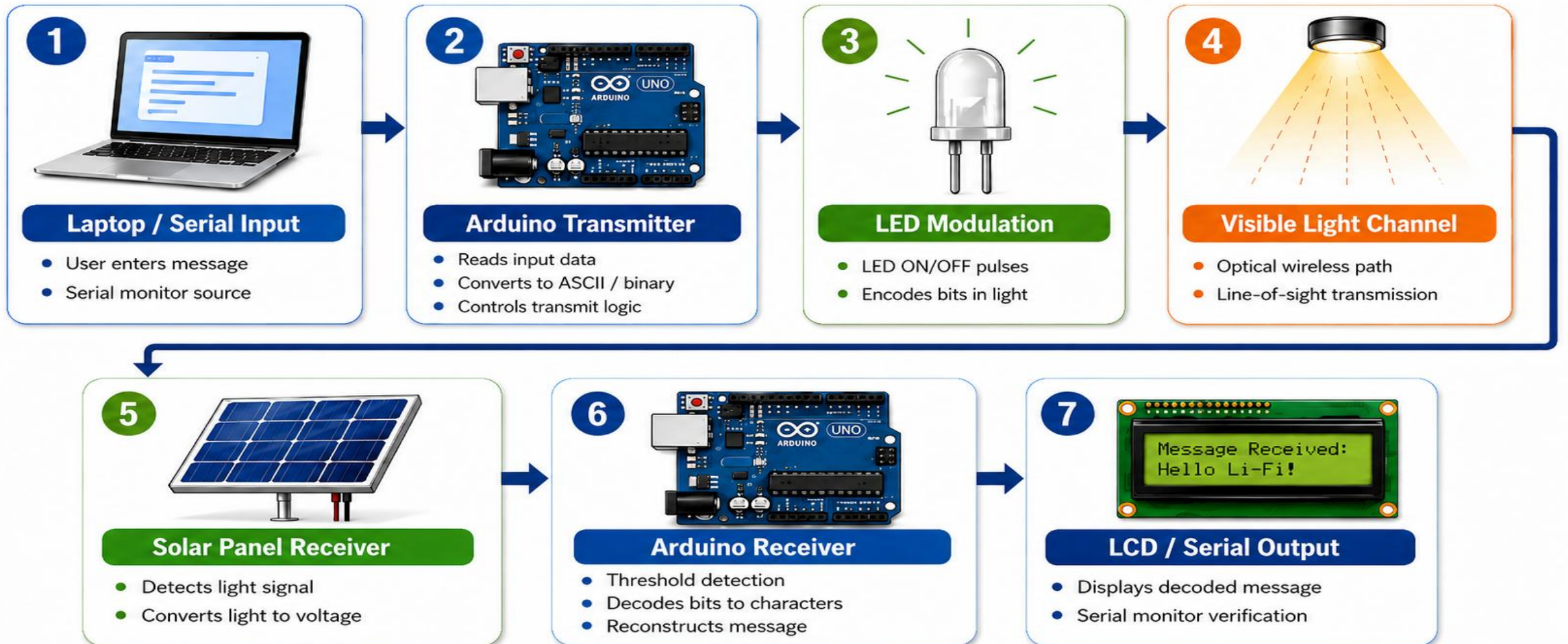


Wiring

System Architecture

End-to-end data flow from input to output through the visible light channel

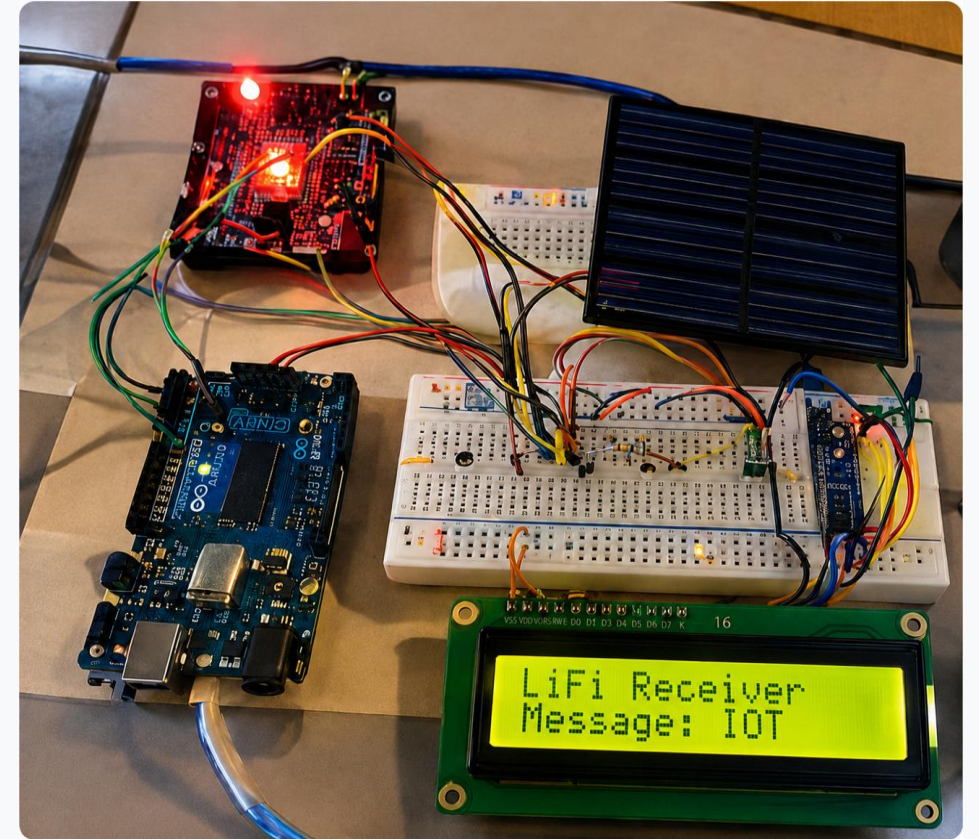
Li-Fi Based Data Transmission – System Architecture



Hardware Components

Essential components for building the Li-Fi prototype system

Arduino Uno	2	Transmitter and receiver boards
LED	1+	Creates light pulses for data transmission
Solar Panel / Photodiode	1	Detects light intensity as receiver sensor
16x2 LCD Display	1	Displays received characters and output
I2C LCD Module	1	Connects LCD using SDA/SCL pins
Resistors	As needed	Protects LED and stabilizes circuit
Breadboard	1	Prototype circuit assembly
Jumper Wires	As needed	Electrical connections
USB Cables	2	Upload code and power Arduinos



Additional Tools

- ✓ Laptop with Arduino IDE
- ✓ Power bank / adapter (optional)
- ✓ Serial Monitor

Wiring & Connections

Detailed circuit connections for transmitter, receiver, and display modules

💡 LED Transmitter Circuit

+ LED positive → Arduino **digital pin 13**

- LED negative → Resistor → **GND**

📡 Solar Panel / Sensor Receiver

+ Positive signal → Arduino **analog pin A3**

- Negative → **GND**

📺 I2C LCD Display Connection

⚡ **VCC** → 5V

— **GND** → GND

↔ **SDA** → A4

🕒 **SCL** → A5

⚙️ Serial Communication

Baud Rate **9600**

📘 Transmitter Arduino uses Serial Monitor at 9600 baud for input. Receiver outputs to LCD display.

Connection Summary

- ✓ LED: Pin 13 + Resistor + GND
- ✓ Sensor: A3 + GND
- ✓ LCD: 5V, GND, A4, A5
- ✓ Serial: 9600 baud

Implementation, Testing & Business Model

Data encoding, transmission logic,
receiver decoding, testing and possible business model



Encoding



Decoding



Testing

Data Encoding & Transmission

Converting characters to binary and transmitting as LED light pulses

Encoding Process



Example: Message "HI" = H(01001000) + I(01001001)

Bit stream: 01001000 01001001

ASCII Character Table

Character A 01000001	Character H 01001000	Character I 01001001	Character O 01001111	Character T 01010100
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</> Transmitter Algorithm

- 1 Read text/character from Serial Monitor
- 2 For each character: convert to ASCII byte
- 3 Loop through 8 bits
- 4 If bit=1: LED ON; If bit=0: LED OFF
- 5 Wait for sampling delay
- 6 Repeat until message complete

Key Parameters

Sampling Delay	Depends on message
Bit Duration	Consistent
Data Format	8-bit ASCII
Loop polling delay	450 ms
Pre-read buffer delay	200 ms

Receiver Decoding Logic

Threshold-based detection and character reconstruction from light signals

Threshold Calibration

LED OFF

~350

Analog Value

Threshold

500

Decision Boundary

LED ON

~600

Analog Value

↓ < 500 = Binary 0

> 500 = Binary 1 ↑

Decoding Process

- 1 Read analog value from solar panel/photodiode
- 2 Compare with threshold (500)
- 3 Classify as bit 1 (if >500) or bit 0 (if <500)
- 4 Store bit in buffer
- 5 After 8 bits: convert to ASCII character
- 6 Display on LCD / Serial Monitor

Receiver Algorithm

Continuous Monitoring

Read analog value from A3 pin

Threshold Decision

if (value > 500) → bit = 1
else → bit = 0

Buffer Management

Collect 8 bits → ASCII conversion

Output Display

Print character to LCD/Serial

Calibration Notes

- ⚠ Values vary with ambient light
- ⚠ Distance affects signal strength
- ⚠ Alignment is critical
- ✓ Threshold: 500 (current setup)

Testing & Calibration Results

Comprehensive validation of binary and character data transmission

T1	0	LED OFF detected	Pass
T2	1	LED ON detected	Pass
T3	HI / IOT	Receiver displays "HI /IOT"	Pass
T4	hello	Receiver displays "hello"	Pass
T5	12345	Receiver displays "12345"	Pass
T6	Looonger message	Not whole message -> short display	Failed
T7	Distance Test	Accuracy vs. range	Tested
T8	Ambient Light	Robustness check	Tested

 **Recommendation:** Keep LED–receiver distance short with direct alignment for optimal performance.

Calibration Values



Test Results Summary

Binary Tests	2/2 Passed
Character and Number Tests	3/4 Passed
External Tests	2/2 Passed

Business Model for Healthcare

Secure light-based data transmission for RF-sensitive environments



Problem: RF Interference

WiFi/RF signals interfere with MRI machines, ICU monitors, and pacemakers. Hospitals require a completely interference free data network to protect patient safety.

✓ Zero RF emission – patient-safe



Solution: LiFi Transmission

LiFi transmits patient data via visible light with zero RF emission. Signals are physically contained to the room delivering inherently secure, interference-free connectivity.

✓ Room-contained – inherently secure



Revenue Model

Hardware: \$199/bed unit

Installation: \$2,000/ward

SaaS maintenance: \$50/month per ward
scalable recurring revenue from day one.

✓ Scalable SaaS + hardware model



Why Us?

Lowest cost LiFi hardware on market. Solar panel receiver eliminates expensive components.

50,000+ hospitals globally 0.1% capture = \$10M+ opportunity.

✓ 0.1% market capture = \$10M+

"The only data network your MRI machine won't fight with."

Results, Challenges & Future

Project outcomes, IoT course alignment,
limitations addressed, and roadmap for improvements



Outcomes



Challenges



Future

Challenges & Solutions

Key obstacles encountered and effective mitigation strategies implemented



Ambient Light Interference

Room light changes sensor readings and affects detection accuracy.

✓ **Solution:** Threshold calibration and controlled lighting environment



LED-Receiver Alignment

Signal weakens if LED does not directly face the receiver sensor.

✓ **Solution:** Direct positioning and alignment during demo



Limited Range

Prototype reliable only at short distances due to LED intensity.

✓ **Solution:** Close-range demo; future: stronger LED/amplifier



Manual Threshold Tuning

Different rooms require different threshold values for optimal detection.

✓ **Solution:** Documented calibration method and procedure



Timing Mismatch

Wrong sampling rate can corrupt bits and characters during transmission.

✓ **Solution:** Fixed sampling delay and stable timing



Low Data Rate

Arduino timing and sensor response limit transmission speed.

✓ **Solution:** Accept as prototype limitation; optimize later



Key Insight

All challenges were successfully addressed through calibration, careful positioning, and documented procedures, enabling reliable character transmission.

Project Success & Future Roadmap

Final Outcome

The project successfully demonstrates an **end-to-end Li-Fi prototype** where data is encoded into LED light pulses, transmitted through visible light, detected by a solar panel/photodiode receiver, decoded by Arduino, and displayed as readable output.

- ✓ Progressed from ON/OFF binary to character transmission
- ✓ Successfully transmitted messages: HI, IOT
- ✓ Validated threshold-based detection approach

Future Improvements

- 1 Photodiode with amplifier for better sensitivity and longer range
- 2 Automatic threshold calibration for changing light conditions
- 3 Manchester encoding or start/stop bits for better synchronization
- 4 Checksum or parity bit for error detection and correction
- 5 Increased data rate through optimized timing and electronics
- 6 Enclosure design to reduce ambient light noise interference

Thank You

Li-Fi Based Data Transmission



Team Members

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